

Barcode Scan Inventory System

Receive, dispatch & track every product by scanning — D-Mart / Walmart style.

Business: K Girdharlal — Restaurant & Catering Supplies **Location:** Surat Store

Status: Live on store network

1 • What this system does

A web-based inventory system where staff manage stock by **scanning barcodes** — no manual data entry for day-to-day work. Every scan instantly updates stock, records who did it and when, and feeds live reports. It works exactly like the checkout and receiving systems at large retail chains.

✓ 523 products loaded & live

✓ Scan-in / Scan-out working

✓ FIFO batch & expiry tracking

✓ Role-based logins

✓ Full audit trail

✓ Live on store WiFi

523

Products in the live catalogue

27

Product categories

3

Scan modes — Setup · Inward · Outward

5

Staff role logins (Admin → View-only)

2 • How it works (the simple idea)

Every product gets its barcode **linked once** to a catalogue item — the same one-time setup every retail chain does. After that, the barcode acts as an instant **lookup key**: scan it, and the system already knows the product, its category, stock, and batches. **Scanning reads the number and looks it up — that is the entire engine behind D-Mart and Walmart too.**

Honest note for review: No system "reads product details out of a barcode" — a barcode is only an ID number. Large stores build a complete catalogue once, then scan against it forever. We have done the same: the catalogue is loaded; the only remaining step is linking each product's barcode to it (one time).

3 · The workflow

One-time setup — link the barcodes

- 1 Open the Scan screen → SETUP mode**
A dedicated mode just for linking — no stock changes.
- 2 Scan a product → pick the matching item (1 click)**
The system shows the catalogue; you confirm which product the barcode belongs to.
- 3 Barcode bonded — forever**
Walk the shelves once. Each product is now permanently recognised.

Faster option: scan all barcodes into one spreadsheet and the system auto-links every product it can identify, leaving only the rest to confirm.

Daily operations — just scan

 SETUP Link barcodes (one-time, per product)	 INWARD Scan to ADD stock when goods arrive	 OUTWARD Scan to REMOVE stock (oldest batch first)
 Goods arrive → INWARD mode → scan Stock goes up. A batch is created with date (and expiry if printed on a carton). Logged under the staff member.		
 Goods leave → OUTWARD mode → scan Stock goes down using FIFO (oldest stock first), so nothing expires sitting on the shelf.		
 Reports update live Current stock, low-stock alerts, expiry alerts, margins and a full audit trail — all automatic.		

4 · What we need to do

- 1

Run the one-time barcode setup
Scan each product once (in the app, or via the bulk spreadsheet) to link its barcode. This is the only manual effort — done once.
- 2

Train staff on the 3 buttons
Setup, Inward, Outward. Takes minutes — it is just "pick a mode and scan."
- 3

Go live
Receive with INWARD, dispatch with OUTWARD. Managers watch reports. That's the daily routine.

5 · Barcode intelligence — what fills in automatically

WHEN YOU SCAN...	WHAT HAPPENS	AUTO?
An already-linked product	Name, category, stock, batches appear instantly; stock moves.	YES
A wholesale GS1 carton	Batch number, expiry date & weight are read straight from the barcode.	YES
A national brand (e.g. Amul)	Looked up online & matched to the catalogue automatically.	YES
A new / local product	Linked once by staff (10 seconds) — then automatic forever.	Once

The system reads **every standard barcode type** — EAN-13, UPC, EAN-8, GS1-128, and weight/price scale labels — and validates each scan.

6 · Staff roles & access

ROLE	CAN DO
Admin	Everything + manage users & setup
Purchase	Purchase orders, vendors, inward
Warehouse	Receive goods, scan inward, view stock
Sales	Dispatch, outward scan, delivery challan
View-only	Read reports only

7 · Access

OPEN ON ANY DEVICE ON THE STORE WIFI

<http://172.16.36.160:5173>

LOGIN	EMAIL	PASSWORD
Admin	admin@fg.local	Admin@123
Warehouse	warehouse@fg.local	Warehouse@123
Sales	sales@fg.local	Sales@123

Demo passwords — to be changed before full rollout.

8 · What's done & what's next

CAPABILITY	STATUS
Product catalogue (523 items, 27 categories)	Live
Scan to receive (Inward) & dispatch (Outward, FIFO)	Live
Universal barcode reader (EAN/UPC/GS1/scale)	Live
Batch, expiry, low-stock & audit tracking	Live
Role logins, reports, delivery challan	Live
One-time barcode setup of the full store	To do (1 session)
Staff training & go-live	Next
Always-on server & daily backup	Planned